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Diabetics Retinopathy Vision Analysis using Image Identification Service Analysis Approach

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Abstract

Retinopathy is one of the leading causes of visual impairment in the developed world, it is provoked by complications of mellitus. The consequence of retina complications are exudate in retina patients perceive no symptoms until visual loss develops. The employment of digital images for diagnosis of eye diseases could be exploited for computerized early detection of exudate from retinal image. In existing System, the blood vessel is detected using morphological operations. To overcome the disadvantages of the existing system the proposed method Exudate Image Identification (EII) approach which will track the images as exudate and non-exudate and it categorize the image with the criteria of mild, moderate and severe based on the captured images. The small exudates are identified with the green channel images. From the images the exudates are calculated with the proposed blood vessel segmentation and optical segmentation algorithm. From this the exudates are calculated with the method called parametric analysis, which can be used for the quick analysis of the exudate evaluation of a Diabetic Retina (DR).

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Keywords: Blood vessel segmentation, optical disc segmentation, parametric analysis

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1. Introduction

In general the middle aged and old age people eye sight may be affected with the cause of the loss of vision. This may occur with the effect of blood vessels in the eyes. The retinopathy may cause because of the high blood sugar which may affect the tiny blood vessels present in the retina. Also the retina infection may affected in the four stages such as mild, moderate, severe and proliferate. As in the mild stage the swelling may cause in the blood vessels, moderate case of infection the retina may got distort and may cause the stoppage of the blood circulation in the blood vessels.

In the severe case the blood circulation may stopped in the retina blood lines and as in the advanced stage a glassy gel may segregated in the eye ball and it fully fill in the eye ball. As of this case the patient vision may get loosed fully. Over identifying the issues and tracking of the eye sight vision with the image cauterisation is a challenging task. Identifying and tracking of images is one of the research topic in image processing [1].

Based on the earlier study the proposed paper discuss an EII approach segmentation of the green channel images[2],[3]. By this approach the exudate images are segregated and analysed with the category of mild, moderate and severe analysis. From that two segmentation are developed and finally executed with the parametric analysis. Based on the two kinds of algorithm a mobile application has been developed and used for the scientific analysis.

2. Literature survey

Alireza Osareh et al (2009) proposed a computational intelligence based approach for detection of exudates in DR images. By this approach the color retinal infected images are segmented with the normalization processing steps. As per this approach the images are classified based on the normalization and segmentation. The image parameters such as color, size and edge strength has been analysed and used for the priority classification using color segmentation algorithm. Also the exudate and non-exudate images are classified and ranked with the feature selection based Genetic algorithm. By this approach the author has proved that the major non-exudate images and majorly used for the requested patients.

Xiaoming Liu ,Tianyu Fu, Zhifang Pan et al (2019) has proposed the automated layer segmentation for OCT (Optical Coherence Tomography) images. By this approach the author has applied the deep convolutional features to classify the OCT image layers. From the classification of the images the analysis can be more comfortable than the analysis of the manual labelling of layer filtering.

Darko Stern , Member, Nicola Giuliani et al (2019) has proposed the method to analyse the age factors from the MRI data. 13 to 25 years of age groups can be taken for the analysis with the forensic practice data. The age categorization analysis can be carried out with the Cropping of Age-Relevant Structures, DCNN Architecture and Visualization of Influence Per Anatomical Site.

M.Alamelu et al (2017), has proposed the new approach called MPU IoT which interconnects the social network and hospital management system. By this method the incurred disease person may identify the disease enquiries by searching through the MPU IoT system, in this system the person who can remotely place can also communicate with the hospital management system. By this way of tracking the disease related queries can provide solutions for disease enquiry.

3. Exudate Image Identification (EII) Approach

The proposed system Exudate Image Identification approach can be categorized based on the collection of exudate affected images. By this proposed system the retina affected images are collected and categorized based on the contrast of the exudate and non-exudate processing[7]. The images are categorized based on the blood vessel segmentation and optical disc segmentation. The three types of classification has been proposed such as

exudate classification, optical disc segmentation and exudate segmentation has been defined for the proposed Exudate Image Identification classification analysis. Figure.1 represents the human eye anatomy view where it shows the representation of the retina, optic nerves, macula, retina blood vessels and so on.[4]

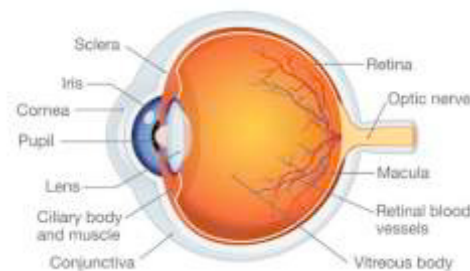


Fig 1 . Human Eye Anatomy

3.1 Exudate Image Identification classification

Exudate Detection

The exudates are segmented by removing blood vessels and optic disc from the green channel image extracted from the fundus image. The steps for exudate detection are as follows.

3.1.1. Blood vessel segmentation

Blood vessels are prone to cause bright lesion like appearance during the segmentation of exudates. Hence it is removed in order to reduce false positive and to improve the accuracy of exudate segmentation. The blood vessels are categorized based on the blood vessel segmentation algorithm. As per this case the blood vessels are monitored with the parameters of ophthalmic artery to the eye[8].

Blood vessel segmentation Algorithm

1. Identify the blood vessels is of the category of mild, moderate and severe.
2. If the case for the mild affected condition analyse the case with the ophthalmic artery flow to the eye. If it exceeds the flow of 30% in the mild case form a cluster as mild affected criteria with the exudate infected ratio in the range of 20% to 30 %
3. If the case for the moderate affected condition analyse the case with the ophthalmic artery flow to the eye. If it exceeds the flow of 50% in the moderate case form a cluster as moderate affected criteria with the exudate infected ratio in the range of 35% to 50 %
4. If the case for the severe affected condition analyse the case with the ophthalmic artery flow to the eye. If it exceeds the flow of 70% in the severe case form a cluster as severe affected criteria with the exudate infected ratio in the range of 55% to 65 %
5. Based on the blood vessel segmentation categorical analysis choose the highly affected case and make it for the image cluster blood vessel segmentation.

Such as the accuracy measure of BVSA = Total number of highest prediction / Total number of the test data.

3.1.2. Optical disc segmentation

The segmentation of optic disc is crucial since it is circular in shape with high contrast and is similar to exudates. optic disc is the point in the eye where it receive millions of neurons. The optical segmentation may carried out by the exudate image identification by means of the effect of the vision loss criteria. Based on the blood vessel segmentation criteria from the collected image the optic disc movement can be calculated with

the symptoms of the person eye vision.

Optical vision segmentation algorithm

1. Vision loss in the eye can be categorized with the starting, middle and full vision loss.
2. The starting stage can be calculated with the optical disc view clearance such as the movements of the flashing in the eye movements. If it is measured less than 20% it can be rated with the starting stage with the infected exudate affected analysis of 20% to 40%
3. The optical disc view clearance is of above 20 and less than 50% it can be rated in the middle stage with the infected exudate affected analysis of 45% to 60 %
4. The optical disc view clearance is of above 70% it can be rated in the full vision stage with the infected exudate affected analysis of above 75%.

Based on this category the optical disc segmentation has been calculated for the collected images.

Accuracy measure of OPVSA = Total number of highest prediction / Total number of the test data.

3.1.3 Exudate segmentation

Exudate segmentation can be carried out by the parametric analysis of the blood vessel segmentation and parametric segmentation. In the parametric analysis the highly infected exudate count ratio can be calculated from the blood vessel segmentation and parametric segmentation.

Exudate segmentation analysis

1. Collect the exudate ratio from the blood vessel segmentation and optical vision segmentation algorithm.
2. Categorize the exudate infection with the category of mild, moderate and severe case.
3. Make the categorization of the exudates with the age factors.
4. If the affected person age less than 40 will comes under category 1.
5. If the affected person above 40 and less than 60 will comes under category 2.
6. If the affected person above 60 and less than 80 will comes under category 3.

Based on the exudate with the person age categorization the final exudate can be rated with respect to the person age.

4. Results and Discussion

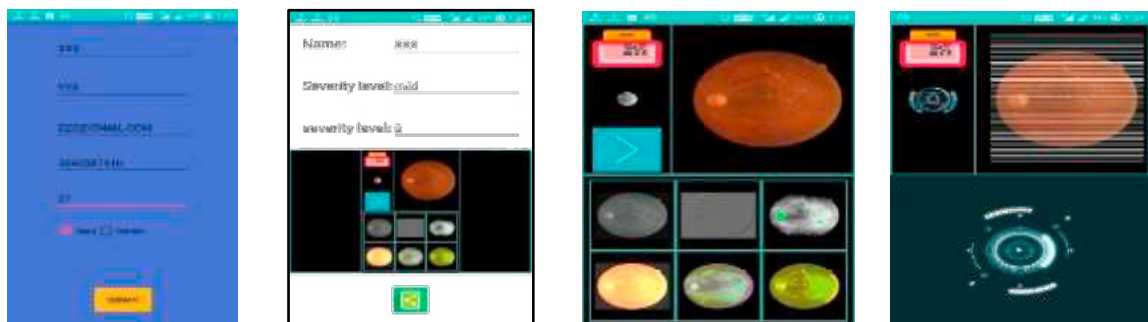


Fig.2 Depicts Retina Affected Pictures

Fig.2 represents the retina infections in the eye vision due to the diabetic disease. The android mobile application was developed with the 100 affected retina images, and are classified with the mild, moderate and severe. The collected images are classified with the blood vessel segmentation and optical disk segmentation algorithm. As of input the affected images in the developed app may categorize the pictures based on the two set of algorithms. The four pictures (1, 2, 3 and 4) presented with the image categorization of the retina affected patients. Pic depicts the mobile android application for doctor's usage. This application can be used for the doctors for the analysis of the patient records with the captured images[9]. Based on the mild, moderate and severe the pictures are categorized and analysed with respect to the blood vessel segmentation algorithm and optical disc segmentation algorithm. Based on the analysis the doctors may come to conclusion to make the analysis of the affected parts.

4.1 performance study analysis

The proposed system image segmentation was compared with the existing algorithms with the methodologies and approaches. Compare to the existing algorithms the proposed system classified with the ophthalmic artery flow and flashing of the eye movement with the mild, moderate and severe category. In the existing system the author [1] was analysed with the categorization of the color retinal images with the category of color segmentation algorithm and fine segmentation based on FCM clustering. With respect to the exudates and non-exudates present in the colour rental images are ranked with the parameters of colour, size and edge strength using Genetic algorithm and in the meanwhile the parameters are classified with the Region level classification of Neural Network (NN).

To reduce more complex analysis of the existing analysis the proposed Exudate Image Identification (EII) Approach was analysed with the blood vessel segmentation algorithm, optical vessel segmentation algorithm and Exudate segmentation algorithms. The proposed system was analysed with the mild, moderate and severe conditions from the exudate affected images. In the existing system tested with the parameters of the colour, size and edge strength with ranking prioritization and classification[10],[11]. The proposed system is analysed with the retinal affected image parameters of the mild, moderate and severe with respect to the analysis of ophthalmic artery flow and flashing of the eye movement of a person. Table.1. represented the comparative study analysis of the existing systems and their approaches, the proposed EII approach can be analysed with the base of this existing works.

Table.1. Study Analysis

S. No	Existing Study Analysis	Approaches used	Advantages
1	Automated Identification of exudate pathology [1]	Segmentation of Color retinal Images using Color Normalization and Contrast Enhancement (with the parameters of size,color,edge strength) 1. color segmentation algorithm 2. Fine segmentation based on FCM clustering 3. Feature Selection Based on GAs	Quick identification of the retinal images and making the ranking and clustering of retinal images using color segmentation, FCM clustering and Genetic algorithms.
2	OCT image classification[2]	Deep residual neural networks and structured random forests.	Have a quick segmentation of the retinal image using the shortest path segmentation
3	Automatic multifactorial age estimation[3]	DCNN architecture Visualization of Influence per Anatomical site	Classification of age from the MRI data

The proposed system was tested with the 50 data set (mobile app) with the affected retinal images and it is to be analysed with the Blood vessel segmentation, optical vision segmentation and exudate segmentation algorithms. The testing analysis can be defined with the mild, moderate and severe count with the count analysis. With the three sort of category the final analysis can be filter out the highly affected exudate data from the given input images.

Table.2 Exudate Infected Analysis

Proposed approaches	Categorical Divisions	Mild	Filtered Count	Moderate	Filtered Count	Severe	Filtered Count	Highly Infected Exudate Analysis
BVSA	ophthalmic artery flow	Exceeds the flow 30%	25	Exceeds the flow 50%	20	Exceeds the flow 70%	15	0.9
	Exudate Infection percentile Ratio	20% to 30 %	6	35% to 50 %	12	55% to 65 %	11	0.46
OPVSA	Flashing of the eye moments	measure d less than 20%	15	optical disc view clearance is of above 20 and less than 50%	16	optical disc view clearance is of above 70%	18	0.68
	Exudate Infection percentile Ratio	20% to 40%	7	45% to 60 %	8	Above 75%.	5	7.5
ESA	Age less than 40	Category 1	20	Category 1	15	Category 1	10	0.7
	above 40 and less than 60	Category 2	07	Category 2	22	Category 2	5	0.54
	above 60 and less than 80	Category 3	16	Category 3	05	Category 3	13	0.58

Table.2 represents the highly affected exudate analysis with respect to the category of the Blood vessel segmentation Algorithm, Optical vision segmentation algorithm and exudate segmentation analysis algorithms. The test analysis was tested with the category of the mild, moderate and the severe case of analysis. In each case the highly affected exudate analysis can be carried out with the high filtered count analysis of the mild, moderate and severe case and it is to be divided with the total number of Input data set. And the exudate infection ratio can be calculated by the mean of the highly infected exudate analysis. The calculated value can be defined in the final exudate analysis. From this the overall accuracy can be calculated with the three measures of the exudate analysis.

From the statistics measure of the above analysis Table. 3 represents the overall highly infected exudate measure with the final accuracy measure. Below table represents the overall infected exudate image ratio analysis with final prediction of the BVSA, OPVSA and ESA analysis. In this case the highly infected exudate analysis and highly infected percentile ratio can be carried out with the three proposed approaches of BVSA, OPVSA and ESA. As of the filtering of data the highly infected rate can be analysed with the mild, moderate and severe category. The final highly infected rate can be measured with the approaches of BVSA and OPVSA comes under less than 40 age category.

Table.3 Overall Highly Infected Rate Analysis

Proposed Approaches	Highly infected Exudate analysis	Highly infected percentile Ratio	Highly Infected rate
BVSA	0.9	0.46	OPVSA
OPVSA	0.68	7.5	
ESA	Category 1(Age less than 40)		

As per the proposed approach a mobile application can be developed with the EII approach. And can be analysed with the three different criteria of age groups of (less than 40, above 40 and less than 60, above 60 and less than 80). From the developed mobile application the exudate infected patient can view the exudate rate analysis and take the respective treatment from the concern doctors. The major limitation of the proposed EII approach it can be used with the limited number of users[5],[6]. As initially the dataset has been tested with 50 input retinal image data with the limitation of 12 users under different age groups. In the future work, the approach can be tested with the more number of users with the more number of data sets.

5. Conclusion

The proposed method Exudate Image Identification (EII) will classify the images with respect to the exudates with the category of images. The images are categorized based on the mild, moderate and severe category of the exudate and with the percentage ratio of the ophthalmic flow range and flash light movement of the eye. By this proposed method the percentage analysis was monitored and measure the accuracy rate with the exudates. By this approach a Mobile Application has been created and used by the retina affected patients and they can take the concern treatment. With extension of this proposed approach the future work can be extended with the number of users.

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